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Hayashi

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(54) **DISPLAY DEVICE**

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H10H 20/831 (2025.01)

H10H 20/857 (2025.01)

(52) **U.S. Cl.**

CPC **H01L 25/0753** (2013.01); **H10H 20/8312** (2025.01); **H10H 20/857** (2025.01)

(58) **Field of Classification Search**

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See application file for complete search history.

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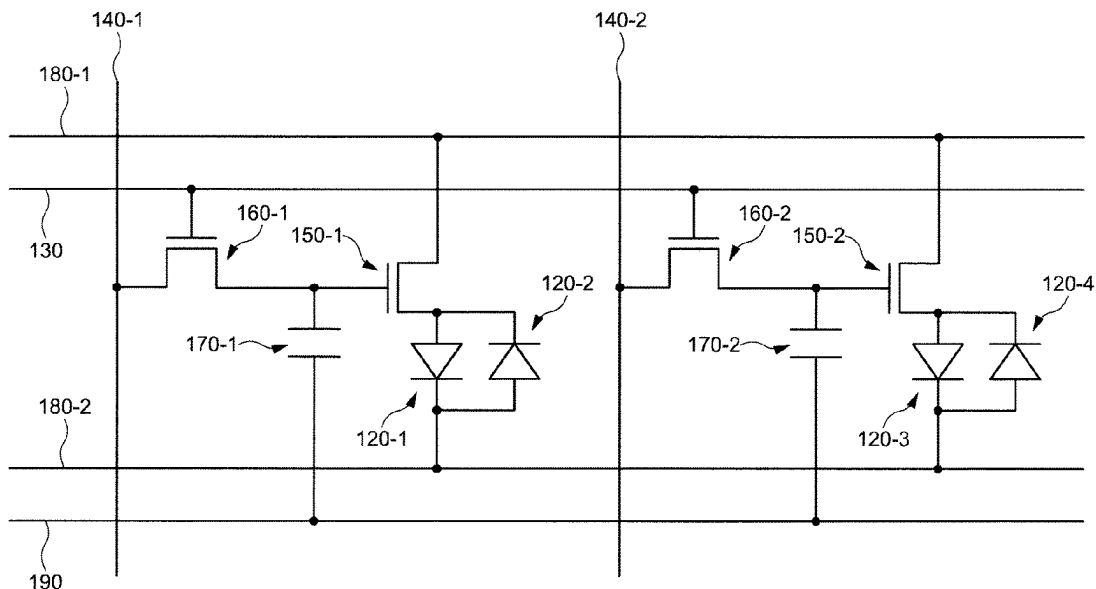
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ABSTRACT

A display device includes a plurality of first pixels and a plurality of second pixels. Each of the plurality of first pixels includes a first LED chip including a first anode and a first cathode, a second LED chip including a second anode and a second cathode, and a first transistor. One of source and drain electrodes of the first transistor is electrically connected to a first power supply voltage line. The other of the source and drain electrodes is electrically connected to the first anode and the second cathode. The first cathode and the second anode are electrically connected to a second power supply voltage line. A first potential of the first power supply voltage line is different to a second potential of the second power supply voltage line.

19 Claims, 10 Drawing Sheets



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FIG. 1

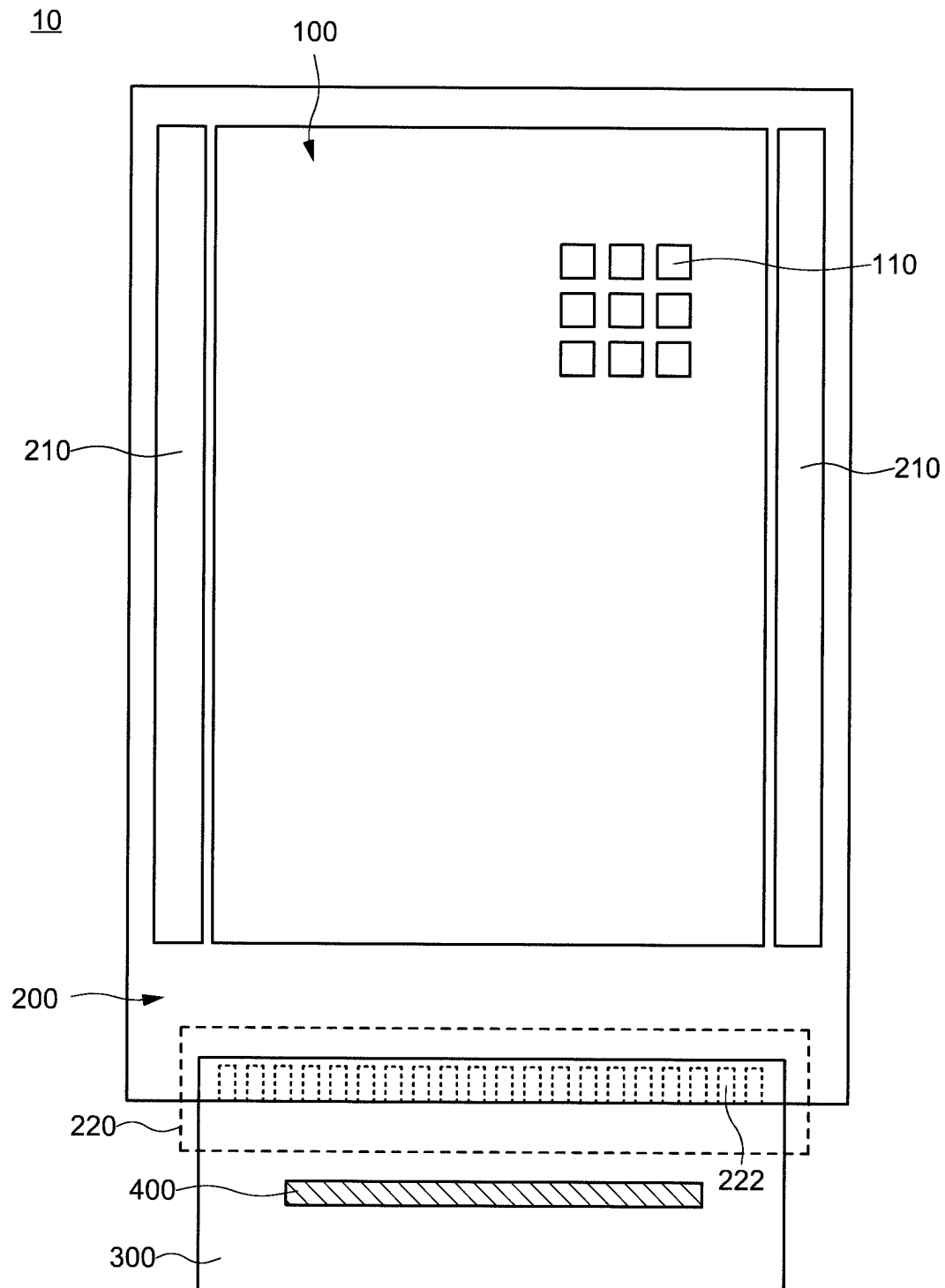


FIG. 2

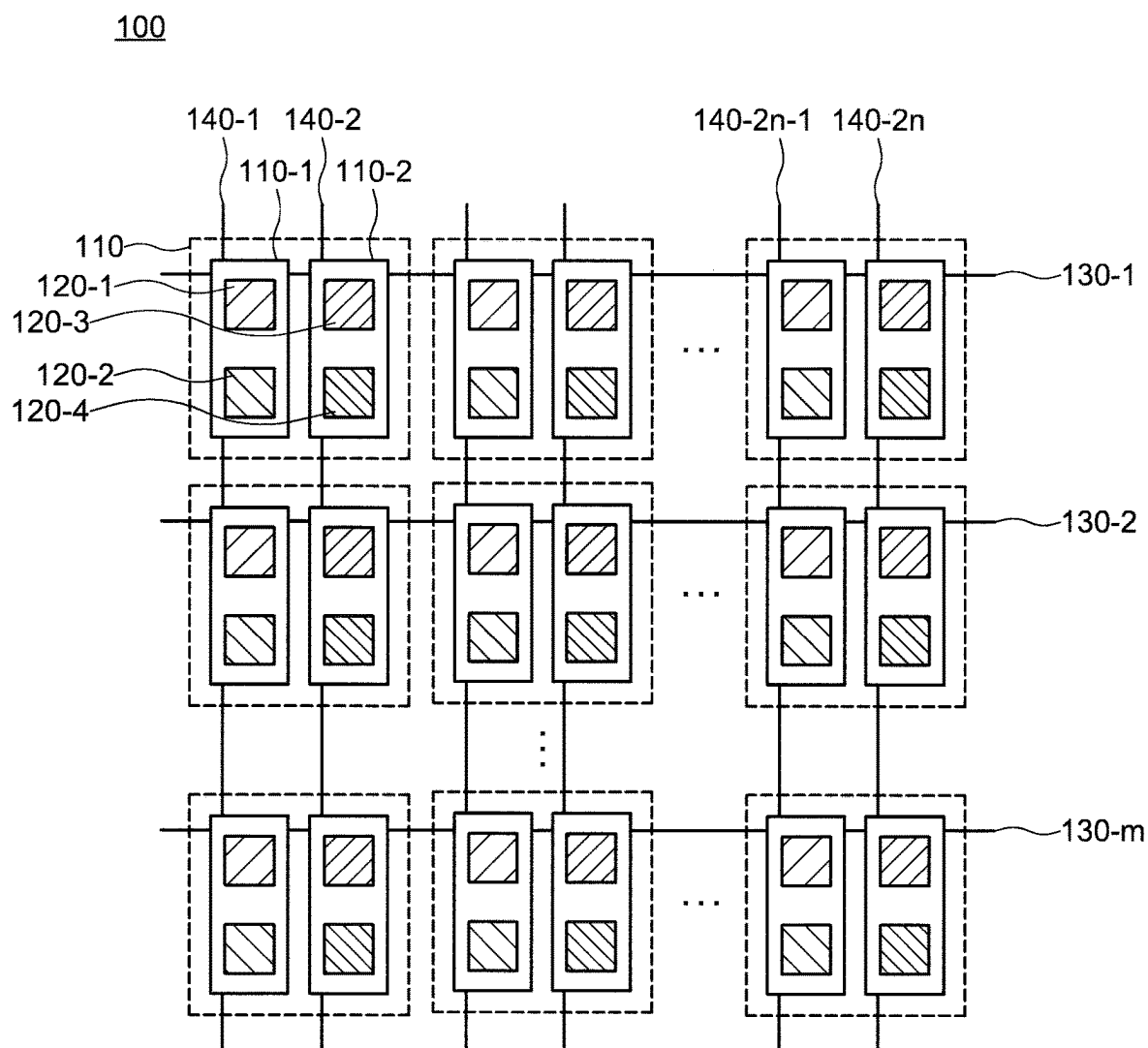


FIG.3

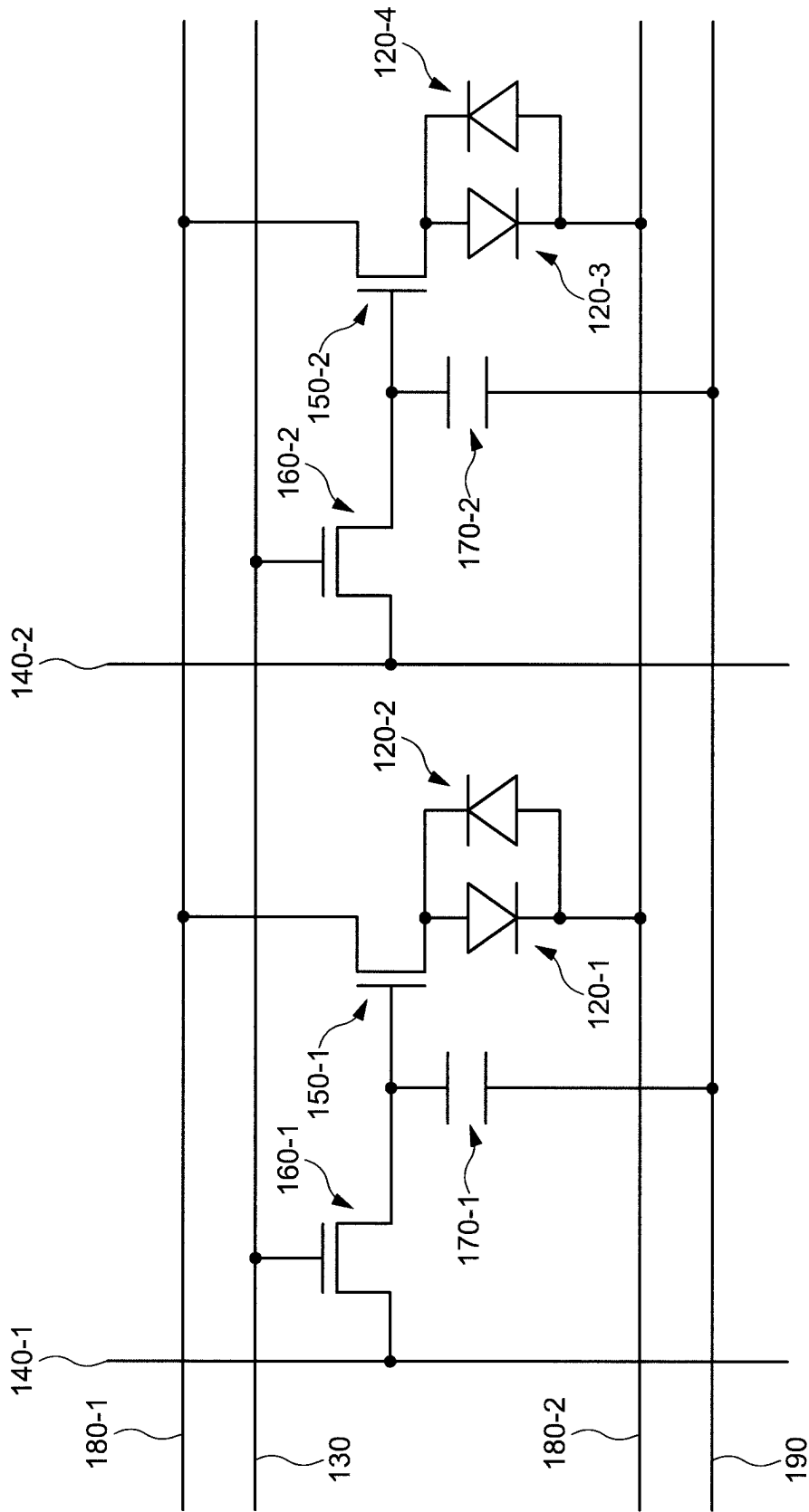


FIG. 4

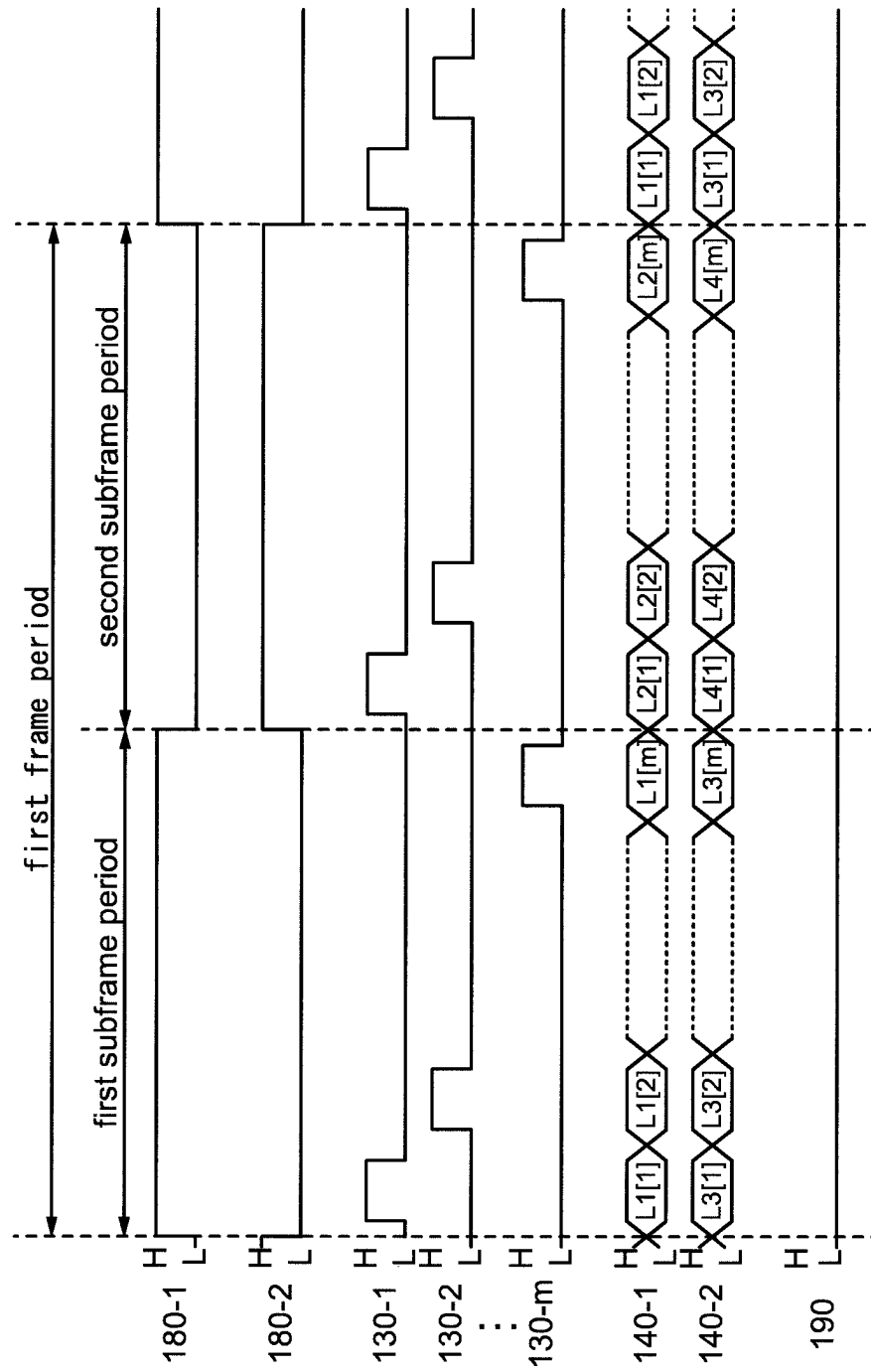


FIG. 5

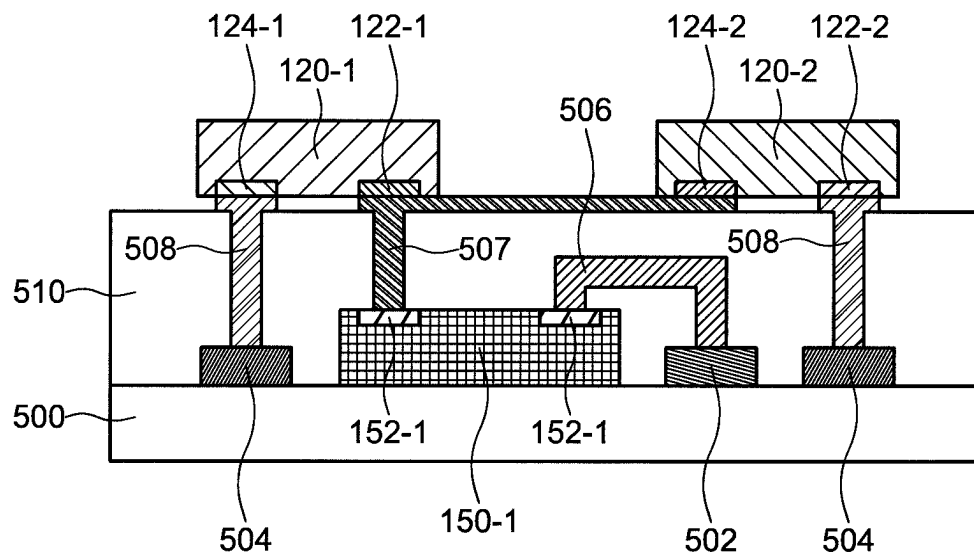


FIG. 6

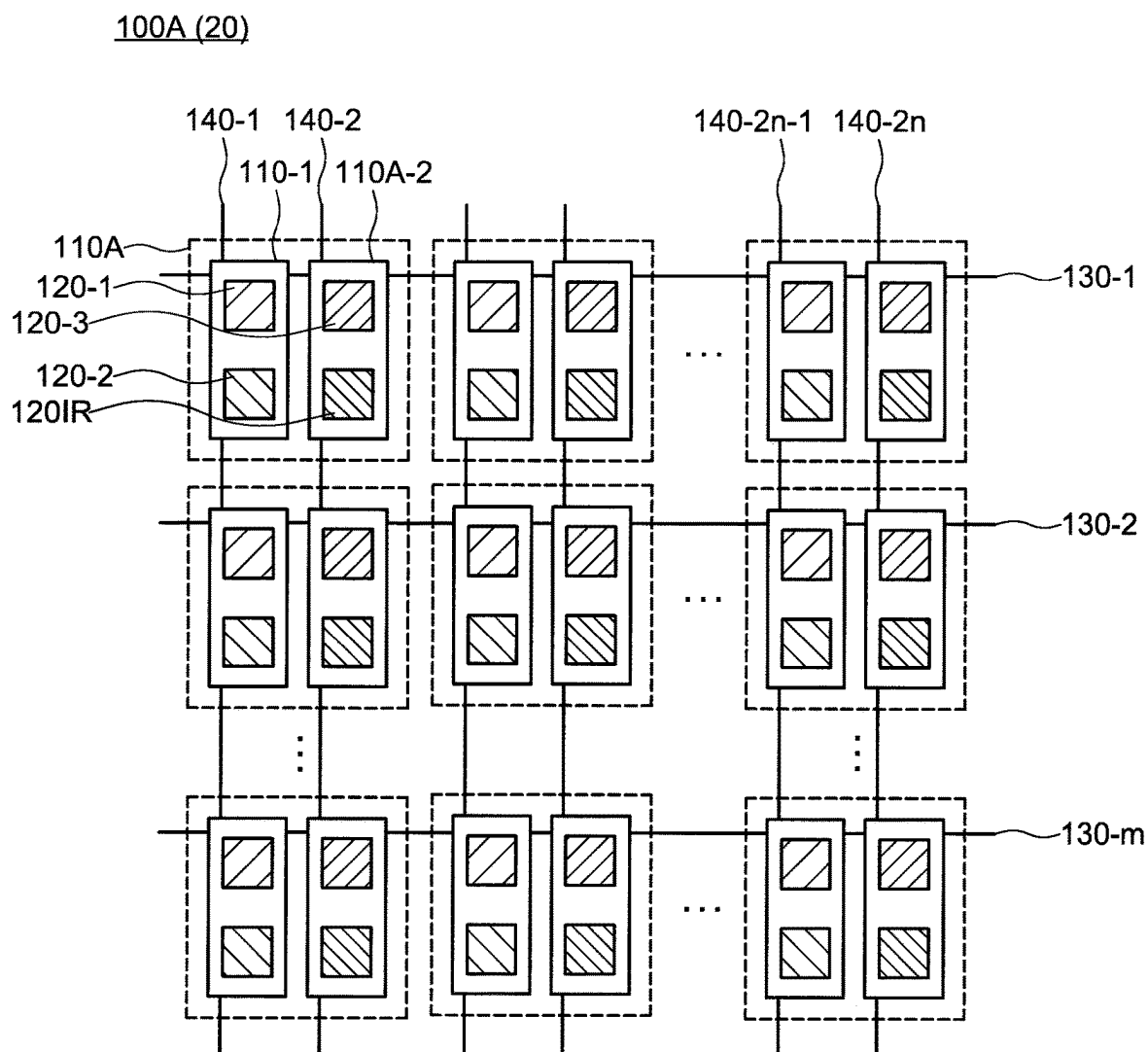


FIG. 7

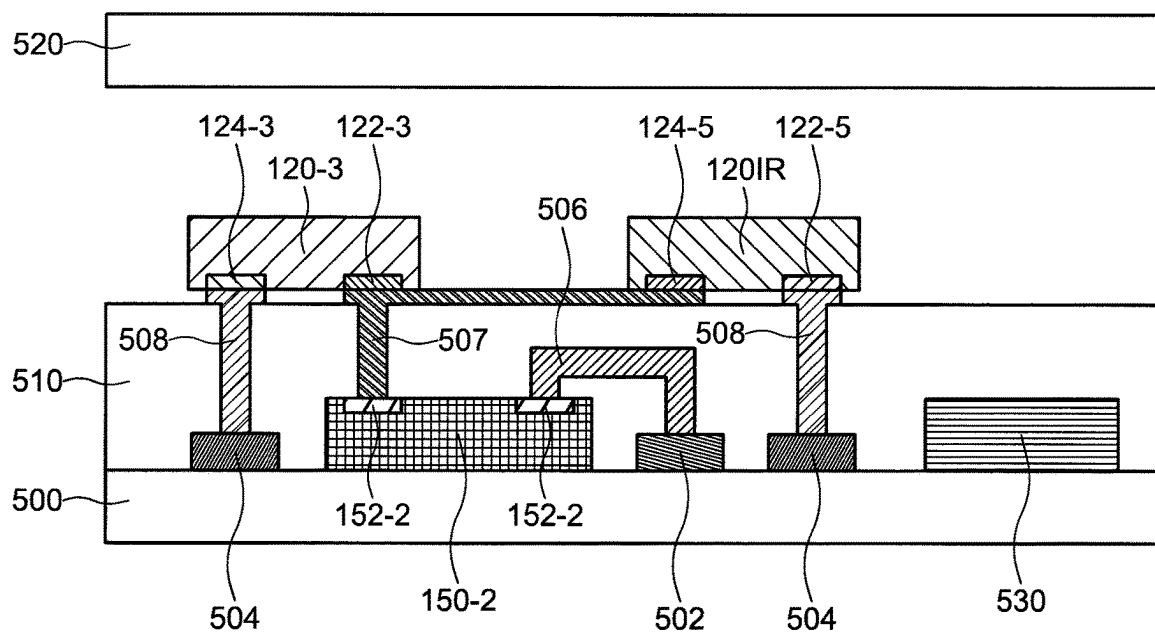


FIG. 8A

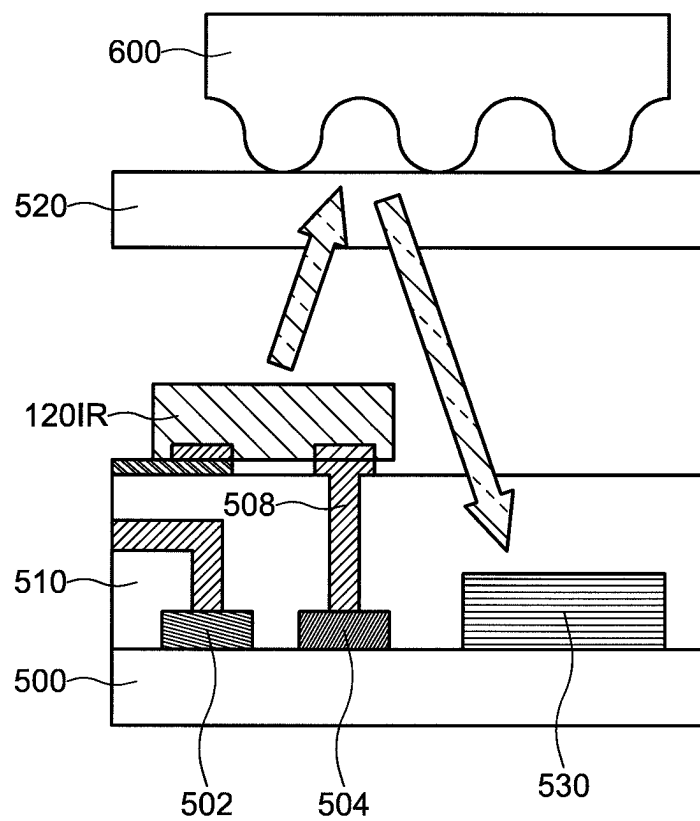


FIG. 8B

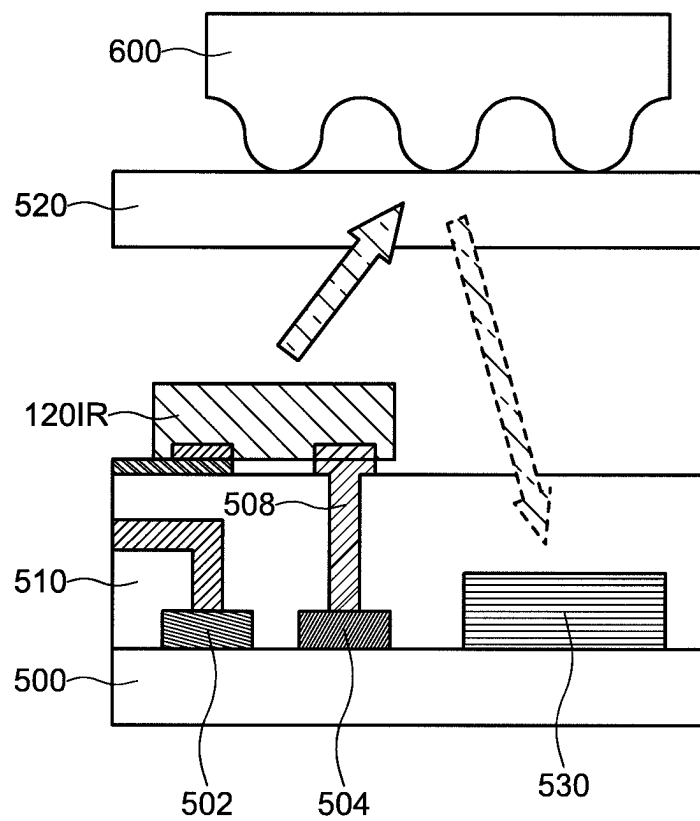
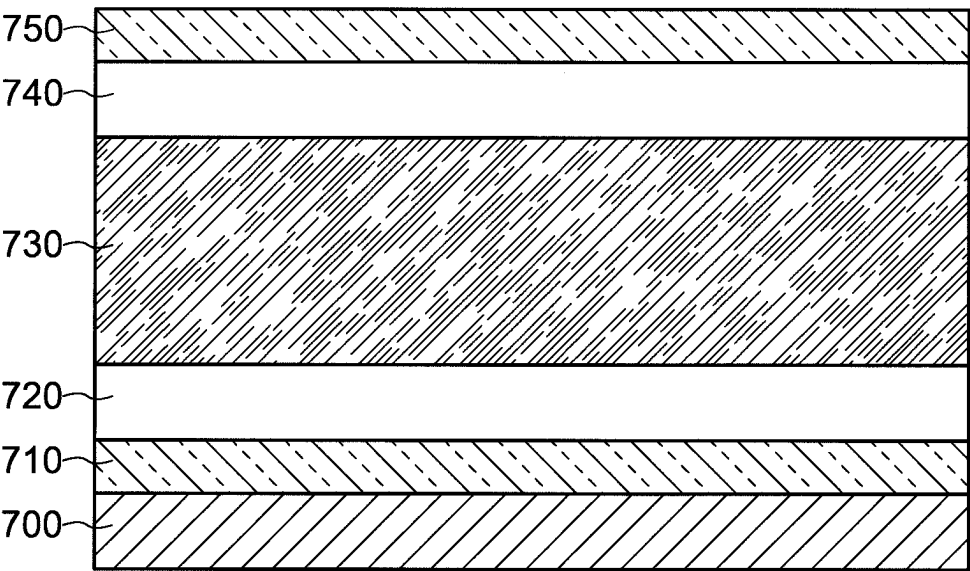


FIG.9

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1 DISPLAY DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of International Patent Application No. PCT/JP2021/001726, filed on Jan. 19, 2021, which claims the benefit of priority to Japanese Patent Application No. 2020-058847, filed on Mar. 27, 2020, the entire contents of which are incorporated herein by reference.

FIELD

One embodiment of the present invention relates to a display device using an LED chip.

BACKGROUND

A liquid crystal display device is a display device that uses liquid crystals and a light source. In the liquid crystal display device, the arrangement of the liquid crystals is changed by applying a voltage to the liquid crystals. The light emitted from the light source is transmitted or shielded due to the difference in the arrangement of the liquid crystals. That is, in the liquid crystal display device, the liquid crystals are used as a switch to control the transmission or non-transmission of the light emitted from the light source.

A light emitting diode (LED) can be used for the light source of the liquid crystal display device. For example, a light source (also referred to as a backlight) in which a plurality of red LED chips, green LED chips, and blue LED chips are arranged in a matrix to obtain white light can be used as a light source of a liquid crystal display device.

However, the LED chip may be damaged by electrostatic discharge due to the application of a voltage in the opposite direction. Therefore, in order to prevent electrostatic discharge damage of the LED chip, a configuration in which a protection diode is connected in parallel to the LED chip is known (see, for example, Japanese Patent Application Laid-Open Publication No. 2008-227423).

On the other hand, focusing on a light emitting element included in the display device, in recent years, a micro LED display device in which a minute LED chip is arranged in each of a plurality of pixels has been developed. The micro LED display device is a self-luminous display device different from the liquid crystal display device, and has excellent visibility. In addition, the LED chip has high emission efficiency and a long life time. Therefore, the micro LED display device is expected to replace the organic EL display device as a next-generation display.

SUMMARY

A display device includes a plurality of first pixels and a plurality of second pixels. Each of the plurality of first pixels includes a first LED chip including a first anode and a first cathode, a second LED chip including a second anode and a second cathode, and a first transistor. One of source and drain electrodes of the first transistor is electrically connected to a first power supply voltage line. The other of the source and drain electrodes is electrically connected to the first anode and the second cathode. The first cathode and the second anode are electrically connected to a second power supply voltage line. A first potential of the first power supply voltage line is different to a second potential of the second power supply voltage line.

2 BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic plan view of a display device according to an embodiment of the present invention.

FIG. 2 is a plan view of an array of pixels in a display portion of a display device according to an embodiment of the present invention.

FIG. 3 is a circuit diagram of a pixel circuit in a pixel of a display device according to an embodiment of the present invention.

FIG. 4 is a timing chart of a pixel circuit of a display device according to an embodiment of the present invention.

FIG. 5 is a cross-sectional view of a pixel of a display device according to an embodiment of the present invention.

FIG. 6 is a plan view of an arrangement of pixels in a display portion of a display device according to an embodiment of the present invention.

FIG. 7 is a cross-sectional view of a display portion of a display device according to an embodiment of the present invention.

FIG. 8A is a diagram illustrating a method of biometric authentication in a display device according to an embodiment of the present invention.

FIG. 8B is a diagram illustrating a method of biometric authentication in a display device according to an embodiment of the present invention.

FIG. 9 is a cross-sectional view of a display device according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

In the light source of Japanese Patent Application Laid-Open Publication No. 2008-227423, since a protection diode needs to be formed separately from the LED chip, manufacturing costs increase. Further, although Japanese Patent Application Laid-Open Publication No. 2008-227423 describes that the LED chip can be used for the protection diode, the LED chip does not directly constitute the light emission of the light source.

One of the objects of an embodiment of the present invention is to provide a display device with a high electrostatic withstand voltage in which an LED chip is used for a light source or a display element.

Each embodiment of the present invention is described below while referring to the drawings. However, the present invention can be implemented in various modes without departing from the gist of the invention and should not be interpreted as being limited to the description of the embodiments exemplified below.

Although the drawings may be schematically represented in terms of width, thickness, shape, and the like of each part as compared with their actual mode in order to make explanation clearer, it is only an example and an interpretation of the present invention is not limited. In addition, in the drawings, the same reference numerals are provided to the same elements as those described previously with reference to preceding figures and repeated explanations may be omitted accordingly.

In the case when a single film is processed to form a plurality of structural bodies, each structural body may have different functions and roles, and the bases formed beneath each structural body may also be different. However, the plurality of structural bodies is derived from films formed in the same layer by the same process and have the same material. Therefore, the plurality of these films is defined as existing in the same layer.

When expressing a mode in which another structure is arranged over a certain structure, in the case where it is simply described as “over”, unless otherwise noted, a case where another structure is arranged directly over a certain structure as if in contact with that structure, and a case where another structure is arranged via another structure over a certain structure, are both included.

In each embodiment of the present invention, as a general rule, the direction from a substrate where a transistor is formed to an LED chip is referred to as “upper” direction and is shown.

First Embodiment

A configuration of a display device **10** according to an embodiment of the present invention is described with reference to FIGS. **1** to **5**.

FIG. **1** is a plan view of the display device **10** according to the embodiment of the present invention.

As shown in FIG. **1**, the display device **10** includes a display region **100** and a peripheral region **200**. The peripheral region **200** is located outside the display region **100**.

The display region **100** is a region for displaying an image or a moving image. Although the shape of the display region **100** shown in FIG. **1** is a rectangle having a long side and a short side, the configuration of the shape of the display region **100** is not limited to this shape. The shape of the display region **100** can be any shape that matches the size or shape of the display device **10**, such as a polygon, a circle, or an ellipse.

The display region **100** includes a plurality of pixels **110**. The plurality of pixels **110** shown in FIG. **1** are arranged in a matrix. However, the configuration of the arrangement of the plurality of pixels **110** is not limited to this arrangement. The plurality of pixels **110** may be arranged in a staggered pattern, for example.

The peripheral region **200** includes a scanning line drive circuit portion **210** and a terminal portion **220**. The scanning line drive circuit portion **210** shown in FIG. **1** is provided along the long side direction of the rectangle of the display region **100**. However, the position of the scanning line drive circuit unit **210** is not limited to this. The scanning line drive circuit portion **210** may be provided, for example, along the short side direction of the rectangle of the display region **100**.

Although the scanning line drive circuit portion **210** shown in FIG. **1** is provided at two locations on the long sides of the rectangle of the display region **100**, the scanning line drive circuit portion **210** may be provided at one location on the long side of the rectangle. Further, the scanning line drive circuit portion **210** may be provided on the short side of the rectangle of the display region **100**.

In the display device **10**, power or a signal can be supplied from the outside through the terminal portion **220**. Therefore, the terminal portion **220** includes a plurality of terminals **222** that can be electrically connected to the outside. The plurality of terminals **222** shown in FIG. **1** are electrically connected to a flexible printed circuit substrate (FPC) **300**. Further, a driver IC **400** is provided on the flexible printed circuit substrate **300**.

The terminal portion **220** is provided at an end of the display device **10**. A video signal and a control signal are supplied to the display device **10** from a controller (not shown) provided outside the display device **10** through the flexible printed circuit substrate **300**. Further, the video signal and the control signal are converted into signals for the display device **10** through the driver IC **400**, and are

input to the plurality of pixels **110** and the scanning line drive circuit portion **210**, respectively. Furthermore, not only the video signal and the control signal, but also power for driving the scanning line drive circuit portion **210**, the driver IC **400**, and the plurality of pixels **110** are supplied to the display device **10**.

FIG. **2** is a plan view of the arrangement of pixels **110** in the display region **100** of the display device **10** according to the embodiment of the present invention.

In the display region **100** shown in FIG. **2**, the plurality of pixels **110** are provided in a matrix of m rows $\times$ n columns (m and n are natural numbers). As shown in FIG. **2**, each of the plurality of pixels **110** includes a first pixel **110-1** and a second pixel **110-2**. The first pixel **110-1** is provided with a first LED chip **120-1** and a second LED chip **120-2**. Further, the second pixel **110-2** is provided with a third LED chip **120-3** and a fourth LED chip **120-4**. Each of the first LED chip **120-1**, the second LED chip **120-2**, the third LED chip **120-3**, and the fourth LED chip **120-4** is, for example, a red LED chip, a green LED chip, a blue LED chip, and a white LED chip. That is, the first pixel **110-1** is a pixel capable of red light emission and green light emission, and the second pixel **110-2** is a pixel capable of blue light emission and white light emission.

In the following description, when the first LED chip **120-1**, the second LED chip **120-2**, the third LED chip **120-3**, and the fourth LED chip **120-4** are not particularly distinguished, they may be described as the LED chip **120** for convenience.

Further, as shown in FIG. **2**, the display region **100** includes a first scanning line **130-1**, a second scanning line **130-2**, and . . . , an m -th scanning line **130- m** . Furthermore, the display region **100** includes a first data line **140-1**, a second data line **140-2**, . . . , a $(2n-1)$ -th data line **140- $2n-1$** , and a $2n$ -th data line **140- $2n$** .

In the following description, when the first scanning line **130-1**, the second scanning line **130-2**, and . . . , the m -th scanning line **130- m** are not particularly distinguished, they are described as a scanning line **130** for convenience. Further, when explaining two adjacent scanning lines **130**, for convenience, the two scanning lines **130** may be described as a first scanning line **130-1** and a second scanning line **130-2**. Furthermore, when the first data line **140-1**, the second data line **140-2**, . . . , the $(2n-1)$ -th data line **140- $2n-1$** , and the second n data line **140- $2n$** are not particularly distinguished, they may be described as a data line **140**, for convenience. In addition, when explaining two adjacent data lines **140** connected to one pixel **110**, for convenience, the two data lines **140** may be described as a first data line **140-1** and a second data line **140-2**.

The scanning line **130** is provided for each pixels **110** in the m row. That is, one scanning line **130** is provided so as to connect to the plurality of pixels **110** arranged in one row. On the other hand, two data lines **140** are provided for each pixels **110** in the n column. That is, the two data lines **140** are provided so as to be commonly connected to the plurality of pixels **110** arranged in one column.

Here, the connection relationship between the first pixel **110-1** and the second pixel **110-2** and the scanning line **130** and the data lines **140** in one pixel **110** is described. The scanning line **130** is provided so as to be commonly connected to the first pixel **110-1** and the second pixel **110-2**. On the other hand, the data lines **140** are provided so that the first data line **140-1** and the second data line **140-2** are connected to the first pixel **110-1** and the second pixel **110-2**, respectively. Therefore, in the pixel **110**, the first LED chip **120-1** or the second LED chip **120-2** provided in the first

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pixel **110-1** can emit light by selecting the scanning line **130** and the first data line **140-1**. Similarly, the third LED chip **120-3** or the fourth LED chip **120-4** provided in the second pixel **110-2** can emit light by selecting the scanning line **130** and the second data line **140-2**.

A pixel circuit for driving the pixel **110** is described with reference to FIG. 3.

FIG. 3 is a circuit diagram of the pixel circuit in the pixel **110** of the display device **10** according to the embodiment of the present invention.

As shown in FIG. 3, the pixel **110** includes the first LED chip **120-1**, the second LED chip **120-2**, the third LED chip **120-3**, the fourth LED chip **120-4**, the scanning line **130**, the first data line **140-1**, the second data line **140-2**, a first drive transistor **150-1**, a second drive transistor **150-2**, a first selection transistor **160-1**, a second selection transistor **160-2**, a first capacitance element **170-1**, a second capacitance element **170-2**, a first power supply voltage line **180-1**, a second power supply voltage line **180-2**, and a capacitance line **190**.

The first pixel **110-1** includes the first LED chip **120-1**, the second LED chip **120-2**, the first drive transistor **150-1**, the first selection transistor **160-1**, and the first capacitive element **170-1**, and the scanning line **130**, the first data line **140-1**, the first power supply voltage line **180-1**, the second power supply voltage line **180-2**, and the capacitance line **190** are connected to the first pixel **110-1**. On the other hand, the second pixel **110-2** includes the third LED chip **120-3**, the fourth LED chip **120-4**, the second drive transistor **150-2**, the second selection transistor **160-2**, and the second capacitance element **170-2**, and the scanning line **130**, the second data line **140-2**, the first power supply voltage line **180-1**, the second power supply voltage line **180-2**, and the capacitance line **190** are connected to the second pixel **110-2**.

Since the drive of the second pixel **110-2** is the same as the drive of the first pixel **110-1**, the connection relationship and the drive in the first pixel **110-1** is described here, and the description of the connection relationship and the drive in the second pixel **110-2** is omitted.

The scanning line **130** is electrically connected to a gate electrode of the first selection transistor **160-1**. One of source and drain electrodes of the first selection transistor **160-1** is electrically connected to the first data line **140-1**, and the other of the source and drain electrodes of the first selection transistor **160-1** is electrically connected to a gate electrode of the first drive transistor **150-1**. One of source and drain electrodes of the first drive transistor **150-1** is electrically connected to the first power supply voltage line **180-1**, and the other of the source and drain electrodes of the first drive transistor **150-1** is electrically connected to an anode of the first LED chip **120-1** and a cathode of the second LED chip **120-2**. A cathode of the first LED chip **120-1** and an anode of the second LED chip **120-2** are electrically connected to the second power supply voltage line **180-2**. One electrode of the first capacitance element **170-1** is electrically connected to the other of the source and drain electrodes of the first selection transistor **160-1**, and the other electrode of the first capacitance element **170-1** is electrically connected to the capacitance line **190**.

When the scanning line **130** is selected, the first selection transistor **160-1** is turned on, and a signal voltage is supplied from the first data line **140-1** to the gate electrode of the first drive transistor **150-1**. Further, the first drive transistor **150-1** is turned on, and a current corresponding to the signal voltage is supplied from the first power supply voltage line

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180-1 or the second power supply voltage line **180-2** to the first LED chip **120-1** or the second LED chip **120-2**.

The first LED chip **120-1** and the second LED chip **120-2** are electrically connected in parallel. However, the first LED chip **120-1** and the second LED chip **120-2** are connected in parallel so that their polarities are opposite to each other. That is, when current is supplied to the first LED chip **120-1**, no current is supplied to the second LED chip **120-2**. Similarly, when current is supplied to the second LED chip **120-2**, no current is supplied to the first LED chip **120-1**.

A first potential of the first power supply voltage line **180-1** is different from a second potential of the second power supply voltage line **180-2**. When the first potential is larger than the second potential, current is supplied only from the first power supply voltage line **180-1** to the first LED chip **120-1**. That is, the first LED chip **120-1** is in a light emitting state, and the second LED chip **120-2** is in a non-light emitting state. On the other hand, when the second potential is larger than the first potential, current is supplied only from the second power supply voltage line **180-2** to the second LED chip **120-2**. That is, the second LED chip **120-2** is in a light emitting state, and the first LED chip **120-1** is in a non-light emitting state. Therefore, one of the LED chips **120** of the two LED chips **120** of the first pixel **110-1** can be in a light emitting state, and the other LED chip **120** can be used as a protection diode.

The switching of the potential between the first power supply voltage line **180-1** and the second power supply voltage line **180-2** may be controlled by a circuit provided on the substrate **500**, or may be controlled by the driver IC **400**.

The drive of the pixel circuit is further described with reference to FIG. 4.

FIG. 4 is a timing chart of the pixel circuit of the display device **10** according to the embodiment of the present invention. FIG. 4 shows each potential of the first scanning line **130-1**, the second scanning line **130-2**, . . . , the m-th scanning line **130-m**, the first data line **140-1**, the second data line **140-2**, the first power supply voltage line **180-1**, the second power supply voltage line **180-2**, and the capacitance line.

The display device **10** includes a first subframe period and a second subframe period in one frame period.

In the first subframe period, the first potential of the first power supply voltage line **180-1** is a high potential ("H" in FIG. 4), and the second potential of the second power supply voltage line **180-2** is a low potential ("L" in FIG. 4). Therefore, during the first subframe period, the first LED chip **120-1** of the first pixel **110-1** and the third LED chip **120-3** of the second pixel **110-2** are in a light emitting state.

When the first scanning line **130-1** is selected ("H" in FIG. 4), light emission from the LED chips **120** of the plurality of pixels **110** arranged in the first row becomes possible. When the second scanning line **130-2** is selected ("H" in FIG. 4), light emission from the LED chips **120** of the plurality of pixels **110** arranged in the second row becomes possible. Similarly, when the m-th scanning line **130-m** is selected ("H" in FIG. 4), light emission from the LED chips **120** of the plurality of pixels **110** arranged in the m-th row becomes possible.

In each of the plurality of pixels **110**, when the first data line **140-1** is selected ("H" in FIG. 4), light emission from the first LED chip **120-1** of the first pixel **110-1** becomes possible ("L1[1]", "L1[2]", . . . , "L1 [m]" in FIG. 4). Further, when the second data line **140-2** is selected ("H" in FIG. 4), light emission from the third LED chip **120-3** of the second pixel **110-2** becomes possible ("L3[1]", "L3[2]", . . .

, “L3[m]” in FIG. 4). That is, the first data line 140-1 and the second data line 140-2 are sequentially selected in each of the plurality of pixels 110.

In the second subframe period, the second potential of the second power supply voltage line 180-2 is a high potential (“H” in FIG. 4), and the first potential of the first power supply voltage line 180-1 is a low potential (“L” in FIG. 4). Therefore, during the second subframe period, the second LED chip 120-2 of the first pixel 110-1 and the fourth LED chip 120-4 of the second pixel 110-2 are in the light emitting state.

When the first scanning line 130-1 is selected (“H” in FIG. 4), light emission from the LED chips 120 of the plurality of pixels 110 arranged in the first row becomes possible. When the second scanning line 130-2 is selected (“H” in FIG. 4), light emission from the LED chips 120 of the plurality of pixels 110 arranged in the second row becomes possible. Similarly, when the m-th scanning line 130-m is selected (“H” in FIG. 4), light emission from the LED chips 120 of the plurality of pixels 110 arranged in the m-th row becomes possible.

In each of the plurality of pixels 110, when the first data line 140-1 is selected (“H” in FIG. 4), light emission from the first LED chip 120-2 of the first pixel 110-1 becomes possible (“L2[1]”, “L2[2]”, . . . , “L2[m]” in FIG. 4). Further, when the second data line 140-2 is selected (“H” in FIG. 4), light emission from the third LED chip 120-4 of the first pixel 110-1 becomes possible (“L4[1]”, “L4[2]”, . . . , “L4[m]” in FIG. 4). That is, the first data line 140-1 and the second data line 140-2 are sequentially selected in each of the plurality of pixels 110.

Therefore, the first LED chip 120-1 and the third LED chip 120-3 can emit light during the first subframe, and the second LED chip 120-2 and the fourth LED chip 120-4 can emit light during the second subframe period. In the display device 10, the first subframe period and the second subframe period are sequentially and repeatedly selected. When the first LED chip 120-1, the second LED chip 120-2, the third LED chip 120-3, and the fourth LED chip 120-4 are a red LED chip, a green LED chip, a blue LED chip, and a white LED chip, respectively, the display device 10 is capable of full-color display by sequentially selecting the first subframe period and the second subframe period.

A configuration of the pixel 110 of the display device 10 is described with reference to FIG. 5.

FIG. 5 is a cross-sectional view of the pixel 110 of the display device 10 according to the embodiment of the present invention. Specifically, FIG. 5 is a cross-sectional view of the first pixel 110-1. Since the configuration of the second pixel 110-2 is the same as the configuration of the first pixel 110-1, the configuration of the first pixel 110-1 is described here, and the description of the configuration of the second pixel 110-2 is omitted.

As shown in FIG. 5, the first pixel 110-1 includes a substrate 500, a first wiring layer 502, a second wiring layer 504, a third wiring layer 506, a fourth wiring layer 507, a fifth wiring layer 508, a protection layer 510, the first LED chip 120-1, the second LED chip 120-2, and the first drive transistor 150-1. The first pixel 110-1 also includes the first selection transistor 160-1 and the first capacitance element 170-1. The first selection transistor 160-1 can have the same configuration as the first drive transistor 150-1, and the first capacitance element 170-1 has a configuration using the insulating layer and electrodes of the first drive transistor 150-1. Therefore, the description of the configurations of the first selection transistor 160-1 and the first capacitance element 170-1 is omitted here.

The first drive transistor 150-1 includes source and drain electrodes 152-1. The first drive transistor 150-1 may be a top gate type transistor or a bottom gate type transistor. Further, the semiconductor layer of the first drive transistor 150-1 may be silicon (amorphous silicon, or polysilicon, etc.), and an oxide semiconductor (zinc oxide (ZnO), gallium oxide (Ga₂O₃), or indium gallium zinc oxide (IGZO)). Further, the first drive transistor 150-1 may be a p-channel type or an n-channel type.

The first LED chip 120-1 includes a first anode 122-1 and a first cathode 124-1. The second LED chip 120-2 includes a second anode 122-2 and a second cathode 124-2. Although the LED chip 120 shown in FIG. 5 has a horizontal LED structure (horizontal electrode structure), the structure of the LED chip 120 is not limited to this structure. The LED chip 120 may have a vertical LED structure (vertical electrode structure).

As shown in FIG. 5, the first wiring layer 502, the second wiring layer 504, and the first drive transistor 150-1 are provided over the substrate 500. Further, the protective layer 510 is provided so as to cover the first wiring layer 502, the second wiring layer 504, and the first drive transistor 150-1. The third wiring layer 506 may be provided in the protective layer 510. Each of the fourth wiring layer 507 and the fifth wiring layer 508 is provided on the opening of the protective layer 510 and the protective layer 510. Each of the first LED chip 120-1 and the second LED chip 120-2 is provided on the third wiring layer 506 and the fourth wiring layer.

The substrate 500 can function as a support substrate that supports each layer provided on the substrate 500. For example, a rigid substrate such as a glass substrate, a quartz substrate, and a sapphire substrate can be used as the substrate 500. When the substrate 500 is a transparent substrate, the display region 100 of the display device 10 has transparency. Further, for example, a flexible substrate such as a polyimide substrate, an acrylic substrate, a siloxane substrate, or a fluororesin substrate can be used as the substrate 500. Impurities may be introduced into the flexible substrate in order to improve the heat resistance of the flexible substrate. When the substrate 500 does not require transparency, for example, a silicon substrate, a silicon carbide substrate, a semiconductor substrate such as a compound semiconductor substrate, or a conductive substrate such as a stainless steel substrate may be used as the substrate 500. The substrate 500 may have a silicon oxide film or a silicon nitride film formed on the above-mentioned substrate.

The first wiring layer 502 can function as a wiring for the first power supply voltage line 180-1. For example, a metal such as aluminum (Al), titanium (Ti), molybdenum (Mo), copper (Cu), or tungsten (W), or an alloy thereof can be used as the material of the first wiring layer 502. Further, the first wiring layer 502 can be a single layer or a stacked layer.

The second wiring layer 504 can function as a wiring for the second power supply voltage line 180-2. The same material as that of the first wiring layer 502 can be used for the second wiring layer 504.

The third wiring layer 506 can function as a wiring for electrically connecting the first wiring layer 502 and one of the source and drain electrodes 152-1 of the first drive transistor 150-1. The same material as that of the first wiring layer 502 can be used for the third wiring layer 506.

The fourth wiring layer 507 can function as a wiring for electrically connecting the other of the source and drain electrodes 152-1 of the first drive transistor 150-1 to the second anode 122-1 of the first LED chip 120-1 and the

second LED chip **120-2**. The same material as that of the first wiring layer **502** can be used for the fourth wiring layer **507**.

The fifth wiring layer **508** can function as a wiring for electrically connecting the second wiring layer **504** to the first cathode **124-1** of the first LED chip **120-1** and the second wiring layer **504** to the second anode **122-2** of the second LED chip **120-2**. The same material as that of the first wiring layer **502** can be used for the fifth wiring layer **508**.

Each of the first LED chip **120-1** and the second LED chip **120-2** can be mounted on the fourth wiring layer **507** and the fifth wiring layer **508** through a conductive adhesive. Therefore, an adhesive layer is provided between the fourth wiring layer **507** and the first anode **122-1** and the second cathode **124-1**, and between the fifth wiring layer **508** and the second anode **122-2**. For example, silver paste or solder can be used for the conductive adhesive. Further, an anisotropic conductive film (ACF) can also be used for the adhesive layer.

The protection layer **510** can protect the first drive transistor **150-1** and flatten steps of the first drive transistor **150-1**. For example, an organic material such as photosensitive acrylic or photosensitive polyimide can be used as the material of the protection layer **510**. Further, for example, an inorganic material such as silicon oxide, silicon nitride, aluminum oxide, or aluminum nitride can be used as the material of the protective layer **510**. Furthermore, the protection layer **510** may be a single layer or a stacked layer. When the protective layer **510** is a stacked layer, a stacked layer in which an inorganic material and an organic material are combined can be used.

As described above, in the display device **10** according to the first embodiment, two LED chips **120** can be driven by using one pixel circuit. Therefore, the number of pixel circuits formed in the display region **100** can be reduced. In addition, the number of pixels in the display region **100** can be increased. Therefore, the display device **10** is a high-definition display device. Further, since the number of pixel circuits formed in the display region **100** can be reduced, the aperture ratio of the display region **100** can be increased. Therefore, since the display region **100** has high transparency, the display device **10** is a display device having high transparency. Furthermore, in the display device **10**, one LED chip **120** of the two LED chips **120** that contribute to white light emission can be in a light emitting state and the other LED chip **120** can be used as a protection diode. Therefore, the display device **10** is a display device having a high electrostatic withstand voltage.

Further, the fourth LED chip **120-4** is not limited to a white LED, and may be an LED that emits any of the colors red, blue, and green. That is, the fourth LED chip **120-4** may be an LED chip that emits the same color as any one of the first LED chip **120-1**, the second LED chip **120-2**, and the third LED chip **120-3**.

Furthermore, the emission colors of the first LED chip **120-1**, the second LED chip **120-2**, the third LED chip **120-3**, and the fourth LED chip **120-4** may be selected in consideration of the color balance of the display device **10**. That is, the first LED chip **120-1**, the second LED chip **120-2**, the third LED chip **120-3**, and the fourth LED chip **120-4** can be any combination of emission colors. For example, the first pixel **110-1** is a pixel capable of emitting red light using two red LED chips **120R**, and the second

pixel **110-2** is a pixel capable of emitting green and blue using a green LED chip **120G** and a blue LED chip **120B**.

Second Embodiment

A configuration of a display device **20** according to an embodiment of the present invention is described with reference to FIGS. **6** to **8B**.

FIG. **6** is a plan view of an arrangement of pixels **110A** in a display region **100A** of the display device **20** according to the embodiment of the present invention. In the following description, the description of the same configuration as the display region **100** of the display device **10** according to the first embodiment is omitted, and the configuration different from the display region **100** is mainly described.

As shown in FIG. **6**, each of the plurality of pixels **110A** includes the first pixel **110-1** and a second pixel **110A-2**. The first pixel **110-1** is provided with the first LED chip **120-1** and the second LED chip **120-2**. Further, the second pixel **110A-2** is provided with the third LED chip **120-3** and an infrared LED chip **1201R**.

Each of the first LED chip **120-1**, the second LED chip **120-2**, and the third LED chip **120-3** is, for example, any one of a red LED chip, a green LED chip, a blue LED chip, and a white LED chip. It is preferable that the pixel **110A** can obtain white light emission by controlling the first LED chip **120-1**, the second LED chip **120-2**, and the third LED chip **120-3**.

Further, as shown in FIG. **6**, the display region **100A** includes the first scanning line **130-1**, the second scanning line **130-2**, and . . . , the m -th scanning line **130- m** . Furthermore, the display region **100A** includes the first data line **140-1**, a second data line **140-2**, . . . , the $(2n-1)$ -th data line **140- $2n-1$** , and the $2n$ -th data line **140- $2n$** .

The scanning line **130** is provided so as to be commonly connected to the first pixel **110-1** and the second pixel **110A-2**. On the other hand, the data line **140** is provided so that the first data line **140-1** and the second data line **140-2** are connected to the first pixel **110-1** and the second pixel **110A-2**, respectively. Therefore, in the pixel **110A**, the first LED chip **120-1** or the second LED chip **120-2** provided in the first pixel **110-1** can emit light by selecting the scanning line **130** and the first data line **140-1**. Similarly, the third LED chip **120-3** or the infrared LED chip **1201R** provided in the second pixel **110A-2** can emit light by selecting the scanning line **130** and the second data line **140-2**.

In the second pixel **110A-2**, the third LED chip **120-3** and the infrared LED chip **1201R** are electrically connected in parallel. However, the third LED chip **120-3** and the infrared LED chip **1201R** are connected in parallel so that their polarities are opposite to each other. That is, when current is supplied to the third LED chip **120-3**, no current is supplied to the infrared LED chip **1201R**. Similarly, when current is supplied to the infrared LED chip **1201R**, no current is supplied to the third LED chip **120-3**.

Although not shown in the diagram, the first potential of the first power supply voltage line **180-1** is different from the second potential of the second power supply voltage line **180-2**. When the first potential is larger than the second potential, current is supplied only to the third LED chip **120-3** from the first power supply voltage line **180-1**. That is, the third LED chip **120-3** is in a light emitting state, and the infrared LED chip **1201R** is in a non-light emitting state. On the other hand, when the second potential is larger than the first potential, current is supplied only to the infrared LED chip **1201R** from the second power supply voltage line **180-2**. That is, the infrared LED chip **1201R** is in a light

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emitting state, and the third LED chip **120-3** is in a non-light emitting state. Therefore, one LED chip **120** of the two LED chips **120** in the second pixel **110A-2** can be in a light emitting state and the other LED chip **120** can be used as a protection diode.

In the second subframe period, the second potential of the second power supply voltage line **180-2** is a high potential, and the first potential of the first power supply voltage line **180-1** is a low potential. Therefore, during the second subframe period, the second LED chip **120-2** in the first pixel **110-1** and the infrared LED chip **1201R** in the second pixel **110A-2** are in a light emitting state. Further, when the second data line **140-2** is selected, light emission from the fourth LED chip **120-4** in the second pixel **110A-2** becomes possible.

A configuration of the pixel **110A** of the display device **20** is described with reference to FIG. 7.

FIG. 7 is a cross-sectional view of the pixel **110A** of the display device **20** according to the embodiment of the present invention. Specifically, FIG. 7 is a cross-sectional view of the second pixel **110A-2**.

As shown in FIG. 7, the second pixel **110A-2** includes the substrate **500**, the first wiring layer **502**, the second wiring layer **504**, the third wiring layer **506**, the fourth wiring layer **507**, the fifth wiring layer **508**, the protection layer **510**, a protective cover **520**, a light receiving element **530**, the third LED chip **120-3**, the infrared LED chip **1201R**, and the second drive transistor **150-2**. The second pixel **110A-1** also includes the second selection transistor **160-2** and the second capacitance element **170-2**. The second selection transistor **160-2** can have the same configuration as the second drive transistor **150-2**, and the second capacitance element **170-2** has a configuration using the insulating layer and electrodes of the second drive transistor **150-2**. Therefore, the description of the configuration of the second selection transistor **160-2** and the second capacitance element **170-2** is omitted here.

The third LED chip **120-3** includes the third anode **122-3** and the third cathode **124-3**. The infrared LED chip **1201R** includes a fifth anode **122-5** and a fifth cathode **124-5**.

As shown in FIG. 7, the first wiring layer **502**, the second wiring layer **504**, the light receiving element **530**, and the second drive transistor **150-2** are provided over the substrate **500**. Further, a protection layer **510** is provided so as to cover the first wiring layer **502**, the second wiring layer **504**, the light receiving element **530**, and the second drive transistor **150-2**. The third wiring layer **506** may be provided in the protection layer **510**. Each of the fourth wiring layer **507** and the fifth wiring layer **508** is provided on the opening of the protection layer **510** and the protective layer **510**. Each of the third LED chip **120-3** and the infrared LED chip **1201R** is provided on the third wiring layer **506** and the fourth wiring layer **508**. The protective cover **520** is provided over the third LED chip **120-3** and the infrared LED chip **1201R**.

The protective cover **520** can protect the third LED chip **120-3** and the infrared LED chip **1201R**. The protective cover **520** can be provided with a gap between the protective cover **520** and the third LED chip **120-3** and the infrared LED chip **1201R**. In this case, the protective cover **520** is preferably adhered to the substrate **500** or the protective layer **510**. Further, the protective cover **520** may be provided by filling an adhesive between the protective cover **520** and the third LED chip **120-3** and the infrared LED chip **1201R**. The same substrate as the substrate **500** can be used as the protective cover **520**. The protective cover **520** preferably has transparency.

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The light receiving element **530** can receive infrared rays and convert them into an electric signal. The light receiving element **530** is preferably a photodiode. For example, a material having a band gap corresponding to the wavelength of the infrared rays such as amorphous silicon can be used as the semiconductor layer of the photodiode.

The display device **20** can perform biometric authentication using the infrared LED chip **1201R** and the light receiving element **530**. Therefore, a method of biometric authentication in the display device **20** is described with reference to FIGS. 8A and 8B.

FIGS. 8A and 8B are diagrams illustrating the method of biometric authentication in the display device **20** according to the embodiment of the present invention. In the following description, although fingerprint authentication is described as an example of biometric authentication in the display device **20**, the biometric authentication method is not limited to this. The biometric authentication in the display device **20** can also be applied to, for example, vein authentication or palm print authentication.

As shown in FIG. 8A, a finger **600**, which is an object of fingerprint authentication, is placed on the protective cover **520** of the display device **20**. The finger **600** has protrusions and recesses on its surface that constitute a fingerprint. Therefore, when the finger **600** is placed on the protective cover **520**, there are convex portions that contact the protective cover **520** and concave portions that do not contact the protective cover **520**.

In the display device **20**, infrared light is emitted from the infrared LED chip **1201R** during the second subframe period. The emitted infrared light passes through the protective cover **520** and is reflected at the interface between the protective cover **520** and the finger **600**. The reflected light is detected by the light receiving element **530** and converted into an electric signal. Here, as shown in FIG. 8A, in the recess of the fingerprint, total reflection occurs at the interface between the protective cover **520** and the recess. Therefore, the intensity of the reflected light is hardly decreased in the concave portions of the fingerprint. On the other hand, as shown in FIG. 8B, in the convex portions of the fingerprint, diffuse reflection occurs at the interface between the protective cover **520** and the convex portions. Infrared light is also absorbed by the veins of the finger **600**. Therefore, the intensity of the reflected light is greatly decreased at the convex portions of the fingerprint. That is, the reflected light becomes stronger in the convex portions of the fingerprint, and the reflected light becomes weaker in the concave portions of the fingerprint. Therefore, the fingerprint authentication of the finger **600** can be performed by detecting the intensity of the reflected light with the light receiving element **530**.

As described above, in the display device **20** according to the second embodiment, two LED chips **120** can be driven by using one pixel circuit. Therefore, the number of pixel circuits formed in the display region **100A** can be reduced. In addition, the number of pixels in the display region **100A** can be increased. Therefore, the display device **20** is a high-definition display device. Further, since the number of pixel circuits formed in the display region **100A** can be reduced, the aperture ratio of the display region **100A** can be increased. Therefore, since the display region **100A** has high transparency, the display device **20** is a display device having high transparency. Furthermore, in the display device **20**, one of the two LED chips **120** can be in a light emitting state and the other LED chip **120** can use as a protection diode. Therefore, the display device **20** is a display device having a high electrostatic withstand voltage.

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Further, the display device **20** is a display device capable of biometric authentication by providing an infrared LED chip **1201R** and a light receiving element **530** in the pixel **110A**. Since the infrared LED chip **1201R** can be mounted in the same manner as the other LEDs **120** and the pixel circuit can be commonly used, manufacturing costs of the display device **20** can be suppressed.

In the second embodiment, although the display device using the infrared LED chip is described, the display device **20** may be provided with the LED chip **120** according to the intended use. That is, the second pixel **110A-2** can be a sensor other than biometric authentication. For example, the LED chip **120** mounted in the second pixel **110A-2** is not limited to the infrared LED chip **1201R**, and can be replaced with invisible light such as an ultraviolet LED chip or a radiation source such as X-rays.

Third Embodiment

A configuration of the display device **30** according to the embodiment of the present invention is described with reference to FIG. **9**.

FIG. **9** is a cross-sectional view of the display device **30** according to the embodiment of the present invention.

As shown in FIG. **9**, the display device **30** includes a light source **700**, a first polarizing plate **710**, a first substrate **720**, a liquid crystal element **730**, a second substrate **740**, and a second polarizing plate **750**. That is, the display device **30** is a liquid crystal display device using the liquid crystal element **730**.

The light source **700** functions as a backlight of the liquid crystal element **730**. The LED chip **120** can be used for the light source **700**. Specifically, the light source **700** can be a light source in which a plurality of red LED chips, green LED chips, blue LED chips, and white LED chips are arranged in a matrix to obtain white light. Further, the red LED chip, the green LED chip, the blue LED chip, and the white LED chip can have the same configuration as the pixel **110** of the display device **10** of the first embodiment. That is, even in the light source **700**, two LED chips **120** can be driven by using one pixel circuit.

Further, the light source **700** may be divided into several sections, each section may be provided with a red LED chip, a green LED chip, a blue LED chip, and a white LED chip, and the LED chips **120** may be controlled in each section. That is, the light source **700** may be provided with one section for a plurality of pixels of the display portion. By having such a configuration of the light source **700**, the number of LED chips **120** provided in the light source **700** can be reduced. Furthermore, since the distance between the LED chips **120** arranged in the light source **700** can be separated, the aperture ratio of the light source **700** becomes high. Therefore, the display device **30** is a display device having high transparency.

Further, the light source **700** may include a lens or a diffuser in order to reduce the luminance nonuniformity.

Although the light source **700** shown in FIG. **9** has a so-called direct light type configuration provided in a flat surface portion of the display device **30**, the configuration of the light source **700** is not limited to this. The light source **700** may have a so-called edge light type configuration provided at the edge of the display device **30**.

The first polarizing plate **710** and the second polarizing plate **750** can polarize the light emitted from the light source **700** in a certain direction. The first polarizing plate **710** and

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the second polarizing plate **750** can be provided with so-called cross Nicols whose transmission axes are orthogonal to each other.

The first substrate **720** and the second substrate **740** can support and protect the liquid crystal element **730**. The same substrate as the substrate **500** can be used as the first substrate **720** and the second substrate **740**. The first substrate **720** and the second substrate **740** are preferably transparent.

The liquid crystal element **730** includes a liquid crystal and an electrode that applies a voltage to the liquid crystal. The liquid crystal element **730** may be a horizontal electric field drive system in which electrodes are provided on one side of the liquid crystal display, or a vertical electric field drive system in which the liquid crystals are sandwiched by the electrodes.

Further, since the liquid crystal element **730** displays in full color, a color filter can be included.

As described above, in the display device **30** according to the third embodiment, two LED chips can be driven by using one pixel circuit in the light source **700**. Further, one LED chip **120** of the two LED chips **120** can be in a light emitting state and the other LED chip **120** can be used as a protection diode. Therefore, the display device **30** is a display device having a high electrostatic withstand voltage.

Each of the embodiments described above as an embodiment of the present invention can be appropriately combined and implemented as long as they do not contradict each other. Additions, deletion, or design changes of constituent elements, or additions, omissions, or changes to conditions of steps as appropriate based on the respective embodiments are also included within the scope of the present invention as long as the gist of the present invention is provided.

Other effects which differ from those brought about by each of the above described embodiments, but which are apparent from the description herein or which can be readily predicted by those skilled in the art, are naturally understood to be brought about by the present invention.

What is claimed is:

1. A display device comprising:

a plurality of first pixels; and
a plurality of second pixels,

wherein each of the plurality of first pixels comprises:

a first LED chip comprising a first anode and a first cathode;

a second LED chip comprising a second anode and a second cathode; and

a first transistor,

one of source and drain electrodes of the first transistor is electrically connected to a first power supply voltage line,

another of the source and drain electrodes is electrically connected to the first anode and the second cathode, the first cathode and the second anode are electrically connected to a second power supply voltage line,

a first potential of the first power supply voltage line is different to a second potential of the second power supply voltage line,

the first LED chip emits a first color, and

the second LED chip emits a second color different from the first color.

2. The display device according to claim 1, wherein the second LED chip is in a non-light emitting state when the first potential is larger than the second potential.

3. The display device according to claim 2, wherein the first LED chip is in a non-light emitting state when the second potential is larger than the first potential.

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4. The display device according to claim 1,
wherein each of the plurality of second pixels comprises:
a third LED chip comprising a third anode and a third
cathode;
a fourth LED chip comprising a fourth anode and a
fourth cathode; and
a second transistor,
the third LED chip and the fourth LED chip are electri-
cally connected in parallel,
one of source and drain electrodes of the second transistor
is electrically connected to the first power supply
voltage line,
another of the source and drain electrodes of the second
transistor is electrically connected to the third anode
and the fourth cathode,
the third cathode and the fourth anode are electrically
connected to the second power supply line,
the third LED chip emits a third color different from the
first color and the second color, and
the fourth LED chip emits a fourth color different from the
first color, the second color, and the third color.
5. The display device according to claim 4, wherein the
fourth LED chip is in a non-light emitting state when the first
potential is larger than the second potential.
6. The display device according to claim 5, wherein the
third LED chip is in a non-light emitting state when the
second potential is larger than the first potential.
7. The display device according to claim 4, wherein a first
subframe period in which the first potential is larger than the
second potential and a second subframe period in which the
second potential is larger than the first potential are sequen-
tially selected.
8. The display device according to claim 7, wherein in the
first subframe period, a first scanning line connected to the
first LED chip and a second scanning line connected to the
third LED chip are sequentially selected.
9. The display device according to claim 8, wherein in the
second subframe period, the first scanning line connected to
the second LED chip and the second scanning line con-
nected to the fourth LED chip are sequentially selected.
10. The display device according to claim 7, wherein in
each of the first subframe period and the second subframe
period, a first data line connected to the first LED chip and
a second data line connected to the second LED chip are
sequentially selected.
11. The display device according to claim 4,
wherein the first LED chip is a red LED chip, and
the second LED chip is a green LED chip.

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12. The display device according to claim 11,
wherein the third LED chip is a blue LED chip, and
the fourth LED chip is a white LED chip.
13. The display device according to claim 4, wherein one
of the first LED chip, the second LED chip, the third LED
chip, and the fourth LED chip is an infrared LED chip.
14. The display device according to claim 1, wherein the
plurality of first pixels and the plurality of second pixels are
arranged in a display portion.
15. The display device according to claim 14, wherein the
display portion has transparency.
16. The display device according to claim 1, further
comprising a display portion comprising a liquid crystal,
wherein the plurality of first pixels and the plurality of
second pixels are arranged in a light source.
17. The display device according to claim 16, wherein the
display portion has transparency.
18. The display device according to claim 1,
wherein each of the plurality of first pixels further com-
prises:
a first capacitance element, and
a capacitance line;
one of electrodes of the first capacitance element is
electrically connected to a gate electrode of the first
transistor, and
another of the electrodes of the first capacitance element
is electrically connected to the capacitance line.
19. The display device according to claim 4,
wherein each of the plurality of first pixels further com-
prises:
a first capacitance element, and
a capacitance line;
each of the plurality of second pixels further comprises:
a second capacitance element, and
the capacitance line;
one of electrodes of the first capacitance element is
electrically connected to a drain electrode of the first
transistor,
another one of the electrodes of the first capacitance
element is electrically connected to the capacitance
line,
one of electrodes of the second capacitance element is
electrically connected to a gate electrode of the second
transistor, and
another of the electrodes of the second capacitance ele-
ment is electrically connected to the capacitance line.

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